Where the Gigabytes Went · IG

U02b · 24 written + 6 pen-first = 30 marks · Name: _____

Questions labelled Ours are authored. Show working and attach units. Unless stated otherwise, count uncompressed pixel/sample data only, excluding headers, metadata and compression. Sound recordings have one channel.

1. Ours: the storage ladder

(a) State the number of bits in a byte and in a nibble. **[2]**

(b) State the number of bytes in 1 KiB. **[1]**

(c) Name the unit equal to 1024 TiB. **[1]**

(d) State how many PiB are in 1 EiB. **[1]**

2. Ours: a game icon

An icon is 256 × 256 pixels at 8 bits per pixel. Calculate its raw size in KiB. **[3]**

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3. Ours: a recorded sound

A 32-second clip uses 16 384 samples per second and 16 bits per sample. Calculate its raw size in KiB. **[3]**

4. Ours: read the depth unit

An image is 400 × 500 pixels at **3 bytes** per pixel. Calculate its raw size in bytes, showing working. **[2]**

5. Ours: change the picture

The icon in Q2 is stored at 16 bits per pixel instead of 8. State its new size in KiB and explain why it changes. **[2]**

6. Ours: one more rung

Convert 2 PiB to TiB. **[1]**

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7. Ours: exam-shape rehearsal · recorded alert

Fresh authored sound-size calculation; use the stated one-channel, uncompressed model.

A 12-second alert uses a sample rate of 24 576 Hz and 16 bits per sample. Calculate its size in KiB. Show your working. **[3]**

8. Ours: spot the unit error

A student calculates an 800 × 600 image at 8 bits per pixel:

> 800 × 600 × 8 = 3 840 000 bits
> 3 840 000 ÷ 1024 = 3750 KiB

(a) Identify the error and give the correct size in KiB, with corrected working. **[3]**

(b) Two tools report the **same byte count** as 1 GB and approximately 0.931 GiB. Explain the different numbers. Does this conversion itself remove any bytes? **[2]**

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9. Ours: predict the file-size machine · pen first [6]

💻 **Write your predictions before running the deck's file-size cell.** Start with width = 512, height = 512 and depth = 16 bits per pixel. The cell computes bits = width × height × depth, then divides by 8 and 1024 to print KiB. These six marks are human marked; no separate Python file is needed.

(a) Calculate the initial output in KiB. Show the conversion from bits. **[2]**

(b) Change only width to 1024. Predict the new output in KiB and explain its relationship to the initial output. **[2]**

(c) Restore width to 512 and change only depth to 8. Predict the output in KiB and explain why. **[2]**

Then run and compare: use the deck cell to check all three predictions. Continue **Karel 07 PotholePatrol and 08 HurdleJumper** as your keyboard companion, outside the 30 marks.

Above and beyond · unmarked. For an image you can inspect, record its dimensions, format and actual file size. Assuming 24 bits per pixel, calculate its raw-pixel estimate. Suggest why the two sizes differ; state what you cannot infer from dimensions alone.

Additional space - label the question number.